1. Use Kepler's laws (conservation of energy and angular momentum) repeatedly. The total energy of a Kepler orbit is $E = -\frac{GM_\odot m}{2a}$ where a is the semi-major axis.

(i) The semi-major axis increases from $r_\oplus$ to $\frac{r_\oplus + r_{♄}}{2}$. Thus the increase in energy:

$$\frac{1}{2}m\left(v_\oplus + \Delta v\right)^2 - \frac{1}{2}m{v_\oplus}^2 = \frac{GM_\odot m}{2r_\oplus} - \frac{GM_\odot m}{r_\oplus + r_{♄}}$$

$$\Delta v = \sqrt{{v_\oplus}^2 + GM_\odot\left(\frac{1}{r_\oplus} - \frac{2}{r_\oplus + r_{♄}}\right)} - v_\oplus = \mathbf{1.03 \times 10^4}\ \textbf{m/s}$$

Note: Considering that $\frac{m{v_\oplus}^2}{r_\oplus} = \frac{GMm}{{r_\oplus}^2}$, we have ${v_\oplus}^2 = \frac{GM}{r_\oplus}$, leading us to the master equation $v = \sqrt{GM_\odot\left(\frac{2}{r_\oplus} - \frac{2}{r_\oplus + r_{♄}}\right)}$, or $\boldsymbol{v} = \sqrt{\boldsymbol{GM_\odot}\left(\frac{\mathbf{2}}{\boldsymbol{r}} - \frac{\mathbf{1}}{\boldsymbol{a}}\right)}$ in general. This works for any position on an elliptical orbit, where r is the distance between the two planets.

(ii) Half a Kepler period, with $a = \frac{r_\oplus + r_{♄}}{2}$:

$$T = \pi\sqrt{\frac{\left(\frac{r_\oplus + r_{♄}}{2}\right)^3}{GM_\odot}} = 191470139\ \text{s} = \mathbf{6.07}\ \textbf{years}$$

(b) The initial semi-major axis is $\frac{r_\oplus + r_\dagger}{2}$. Since $T \propto a^{3/2}$, so the final semi-major axis is $2^{2/3}r_\dagger$. Lets go through the motion step-by-step:
Firstly, the new velocity when sent to Venus via Hohmann transfer is

$$v_1 = \sqrt{GM_\odot\left(\frac{2}{r_\oplus} - \frac{2}{r_\oplus + r_\dagger}\right)} = 27218.2907\ \text{m/s}$$

Then, when it intersects Venus' orbit for the first assist, by conservation of angular momentum, its velocity is

$$v_2 = \frac{r_\oplus}{r_\dagger}v_0 = 37803.18153\ \text{m/s}$$

Alternatively, this can be taken from $v_2 = \sqrt{GM_\odot\left(\frac{2}{r_\dagger} - \frac{2}{r_\oplus + r_\dagger}\right)}$

The speed needed to transit to the new orbit of $a = 2^{2/3}r_\dagger$

$$v_3 = \sqrt{GM_\odot\left(\frac{2}{r_\dagger} - \frac{1}{2^{2/3}r_\dagger}\right)}$$

Thus the change in velocity $\Delta v = \sqrt{GM_\odot\left(\frac{2}{r_\dagger} - \frac{1}{2^{2/3}r_\dagger}\right)} - \frac{r_\oplus}{r_\dagger}\sqrt{GM_\odot\left(\frac{2}{r_\oplus} - \frac{2}{r_\oplus + r_\dagger}\right)} =$ $3230.841378 \approx \boldsymbol{3230 m/s}$

Note: this can also be found from doing conservation of energy,
$\frac{1}{2}mv_3^2 - \frac{1}{2}mv_2^2 = -\frac{GMm}{2*2^{2/3}r_\dagger} - \left(-\frac{GMm}{r_\oplus + r_\dagger}\right)$

2. Conservation of angular momentum about point on ground:

$$L = mv_0R + I\omega_0 = mv_fR + I\omega_f = \left(mR + \frac{I}{R}\right)v_f$$

The time taken can be computed as

$$T = \frac{|v_f - v_0|}{a}$$

where acceleration/deceleration $a = \frac{F_f}{m} = \frac{\mu_k mg}{m} = \mu_k g$ and

$$v_f - v_0 = \frac{mv_0R + I\omega_0}{mR + \frac{I}{R}} - v_0 = \frac{I}{mR^2 + I}(R\omega_0 - v_0)$$

In the case of part (a),

$$T_a = \boldsymbol{\frac{I}{(mR^2 + I)\mu_k g}R\omega_0}, \qquad v_f = \frac{I\omega_0}{mR + \frac{I}{R}} = \boldsymbol{\frac{I}{mR^2 + I}R\omega_0}$$

In the case of part (b),

$$T_a = \boldsymbol{\frac{I}{(mR^2 + I)\mu_k g}|R\omega_0 - v_0|}, \qquad v_f = \frac{mv_0R + I\omega_0}{mR + \frac{I}{R}} = \boldsymbol{\frac{mv_0R^2 + IR\omega_0}{mR^2 + I}}$$

(c) Similar derivation as above, but replace ω_0 with $-\omega_0$.

Case I: $mv_0R > I\omega_0$. At equilibrium, wheel rolls without slipping to the right with speed $\boldsymbol{\frac{mv_0R^2 - IR\omega_0}{mR^2+I}}$ and angular speed (clockwise) $\boldsymbol{\frac{mv_0R - I\omega_0}{mR^2+I}}$.
Case II: $mv_0R < I\omega_0$. At equilibrium, wheel rolls without slipping to the left with speed $\boldsymbol{\frac{IR\omega_0 - mv_0R^2}{mR^2+I}}$ and angular velocity (anticlockwise) $\boldsymbol{\frac{I\omega_0 - mv_0R}{mR^2+I}}$.
Case III: $mv_0R = I\omega_0$. At equilibrium, wheel is stationary and non-rotating.

For all 3 cases, the wheel takes time $T = \frac{I}{(mR^2+I)\mu_k g}(v_0 + R\omega_0)$ to reach its equilibrium state. During the time T, it rolls with slipping (kinetic friction providing both consant torque and acceleration).

Note: it is much easier to define a direction (e.g. right positive for v, clockwise positive for ω), then your answer will stay the same. You still have to describe the cases and explain why (which is about the total angular momentum being positive, negative, or 0).

3. Voltage is undefined – question was nullified that year. Current is defined.

Let $I_{AB1}, I_{BC1}, I_{CA1}, I_{AB2}, I_{BC2}, I_{CA2}$ be the currents through the wires, all directed clockwise, and those labelled 1 are on the circle while those labelled 2 are on the triangle. Apply Faraday's Law for 4 loops, and Kirchoff's current law:

$$\begin{cases} I_{AB2}r_2 + I_{BC2}(2r_2) + I_{CA2}r_2 = \dfrac{3\sqrt{3}}{4}a^2k \\ I_{AB1}r_1 - I_{AB2}r_2 = I_{BC1}r_1 - I_{BC2}(2r_2) = I_{CA1}r_1 - I_{CA2}r_2 = \dfrac{\pi - \frac{3\sqrt{3}}{4}}{3}a^2k \\ I_{AB1} + I_{AB2} = I_{BC1} + I_{BC2} = I_{CA1} + I_{CA2} \end{cases}$$

Six equations for six unknowns, is solvable. First by symmetry note $I_{AB1} = I_{CA1}$ and $I_{AB2} = I_{CA2}$. Also, we can substitute in $r_1 = \frac{3}{2}r_2$.

$$\begin{cases} I_{AB2} + I_{BC2} = \dfrac{3\sqrt{3}}{8}\dfrac{a^2k}{r_2} \\ 3I_{AB1} - 2I_{AB2} = 3I_{BC1} - 4I_{BC2} = \dfrac{2\left(\pi - \frac{3\sqrt{3}}{4}\right)}{3}\dfrac{a^2k}{r_2} \\ I_{AB1} + I_{AB2} = I_{BC1} + I_{BC2} \end{cases}$$

Combining the last two equations: $5I_{AB2} = 7I_{BC2}$. Substituting into first eqn:

$$\begin{cases} I_{AB2} = I_{CA2} = \dfrac{7\sqrt{3}}{32}\dfrac{a^2k}{r_2} \\ I_{BC2} = \dfrac{5\sqrt{3}}{32}\dfrac{a^2k}{r_2} \end{cases}$$

Using second equation:

$$\begin{cases} I_{AB1} = I_{CA1} = \dfrac{1}{3}\left(\dfrac{2\left(\pi - \frac{3\sqrt{3}}{4}\right)}{3} + \dfrac{7\sqrt{3}}{16}\right)\dfrac{a^2k}{r_2} = \dfrac{a^2k}{3r_2}\left(\dfrac{2\pi}{3} - \dfrac{\sqrt{3}}{16}\right) \\ I_{BC1} = \dfrac{1}{3}\left(\dfrac{2\left(\pi - \frac{3\sqrt{3}}{4}\right)}{3} + \dfrac{5\sqrt{3}}{8}\right)\dfrac{a^2k}{r_2} = \dfrac{a^2k}{3r_2}\left(\dfrac{2\pi}{3} + \dfrac{\sqrt{3}}{8}\right) \end{cases}$$

We have thus solved for all currents. Potential difference is not well-defined.

Note: Because of changes in magnetic field, $\nabla \times E = -\frac{\partial B}{\partial t} \neq 0$, **E is not conservative**. Hence, no unique potential energy function (here, voltage) exists. But current, which is from $E = \sigma J$, exists.

4. The acceleration of the system is $a = \frac{m_2 - m_1}{m_2 + m_1} g = \frac{m}{2M+m} g$. The velocity when the small mass is removed is $v = \sqrt{2ah}$ and after $t = 1$ s, it covers distance $H = vt$. Thus

$$H = t\sqrt{2\frac{m}{2M+m}gh} \therefore g = \frac{2M+m}{2mh}\left(\frac{H}{t}\right)^2 = \frac{2+0.01}{0.02}\left(\frac{0.312}{1}\right)^2 = \mathbf{9.78\ m/s^2}$$

5. (a) Expression for Q indicates that initial Thorium particle is stationary (no kinetic energy of Thorium is present in the expression). Therefore by conservation of momentum,

$$m_{Ra}v_{Ra} + m_\alpha v_\alpha = 0$$

so ratio of kinetic energy of radium to alpha particle is

$$\frac{1}{2}m_{Ra}v_{Ra}^2 : \frac{1}{2}m_\alpha v_\alpha^2 = m_\alpha : m_{Ra}$$

therefore the percentage of Q carried by the alpha particle is

$$\boldsymbol{\frac{m_{Ra}}{m_\alpha + m_{Ra}}}$$

which we can numerically estimate to be

$$\frac{224}{4+224} = \mathbf{0.982}$$

(b) The difference in Qs are

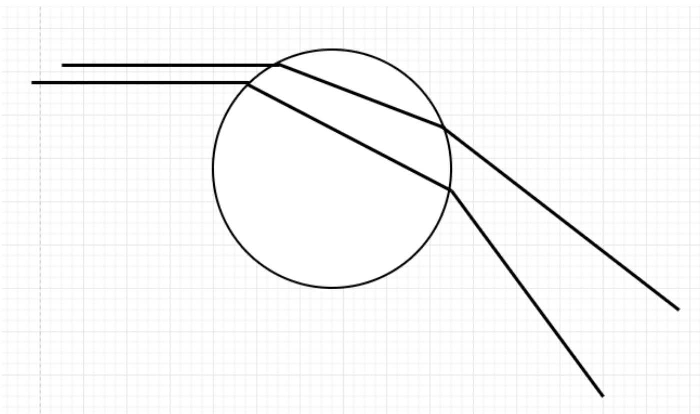
$$\frac{4+224}{224}(5.423 - 5.341) = 0.08346 \text{ MeV}$$

So the energy of first excited state of Ra-224 is

$$0.08346 \text{ MeV} = \mathbf{83.5\ keV}$$

6. For online explanations, search “zeroth order rainbow”.

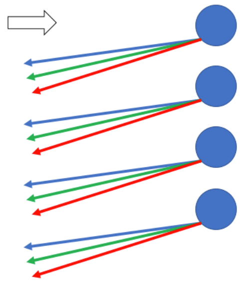
(a) See the diagram:



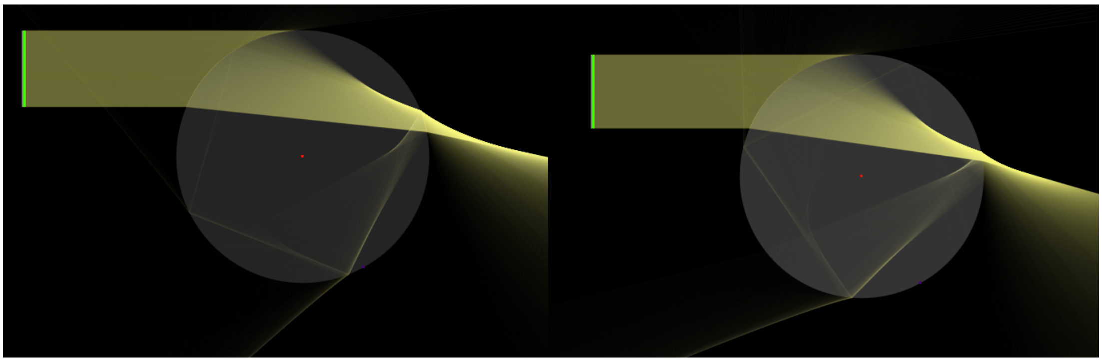
The angles θ_i and θ_r are not labelled, but we know that at each boundary the light is bent by $\theta_i - \theta_r$. So total angle of deviation is

$$\alpha = 2(\theta_i - \theta_r)$$

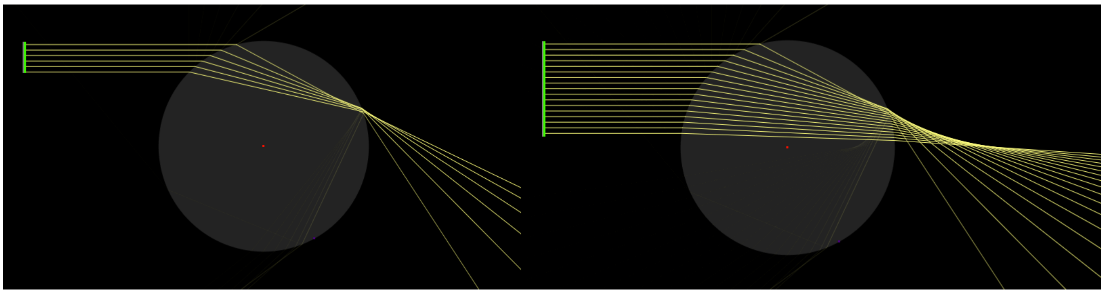
(b) You are seeing all the parallel rays from the sun that hit droplets and refract into your eyes with the specific angle of deviation α. Suppose as incidence angle θ_i, the angle of deviation α reaches a maximum α_{max}. Then α_{max} is in terms of n. Since n varies for light of different wavelengths/frequencies, thus α_{max} will vary across the visible light range; thus at some angles, you no longer see some colours! That is why you see a rainbow.



Lower n – red Higher n – blue



In this diagram, many rays of different θ_i are introduced to the circle, but there is a cluster near the bottom of the image – this is the stationary point, many rays of similar θ_i have the same angle of deviation α, and exit the rainbow together after one . For a different refractive index, this angle will be slightly different. This means that an ordinary rainbow displays rays of different colour at different angles.



On the other hand, if the angle of deviation α increases with θ_i, then there is simply no rainbow (every angle you look at will have light from the whole visible spectrum, which appears white). In the two black images above, introducing more θ_i just results in more output, there is no concentration or common (shared) value of α where rays can concentrate, none of them are even parallel. Hence, there is no isolated colour-based output angle like the previous case.

Ultimately the condition for a rainbow to occur is a stationary point $\frac{d\alpha}{d\theta_i} = 0$.

Let's check for a zeroth order rainbow.

$$\sin\theta_i = n\sin\theta_r \therefore \cos\theta_i \, d\theta_i = n\cos\theta_r \, d\theta_r$$

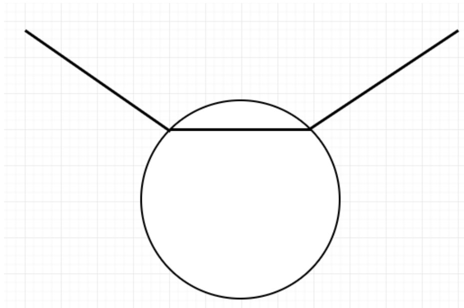
Thus (and we implicitly use $\theta_i > \theta_r \therefore \cos\theta_i < \cos\theta_r$)

$$\frac{d\alpha}{d\theta_i} = 2\left(1 - \frac{d\theta_r}{d\theta_i}\right) = 2\left(1 - \frac{\cos\theta_i}{n\cos\theta_r}\right) > 2\left(1 - \frac{1}{n}\right) > 0$$

so there is no stationary point, so there is no zeroth order rainbow.

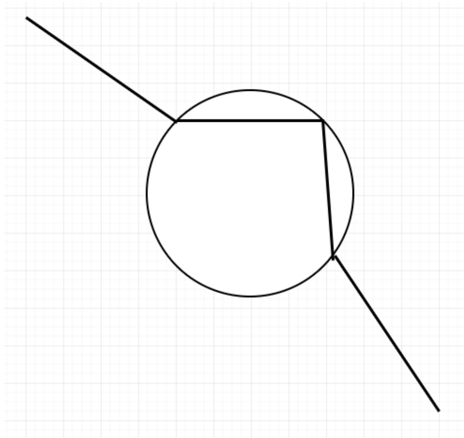
<u>It is a good exercise to check that there is a first order rainbow, and that you'll see it at $\approx 40° - 42°$ (taking $n \sim 1.33$).</u>

(c) Zeroth order:



$$\alpha = 2(\theta_r - \theta_i)$$

First order:



$$\alpha = 2\left(\frac{\pi - 2\theta_r}{2} - (\theta_r - \theta_i)\right) = \pi - 4\theta_r + 2\theta_i$$

(d) Lets do the same thing, finding stationary point. Let's check for a zeroth order bubblebow.

$$n\sin\theta_i = \sin\theta_r \therefore n\cos\theta_i \, d\theta_i = \cos\theta_r \, d\theta_r$$

Thus (and we implicitly use $\theta_i < \theta_r \therefore \cos\theta_i > \cos\theta_r$)

$$\frac{d\alpha}{d\theta_i} = 2\left(\frac{d\theta_r}{d\theta_i} - 1\right) = 2\left(n\frac{\cos\theta_i}{\cos\theta_r} - 1\right) > 2(n-1) > 0$$

so there is no stationary point, so there is no zeroth order bubblebow.

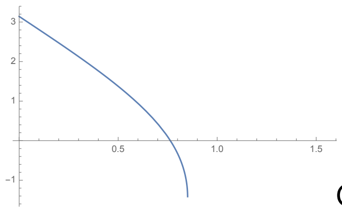
Let's check for a first order bubblebow.

$$\frac{d\alpha}{d\theta_i} = 2 - 4\frac{d\theta_r}{d\theta_i} = 2 - 4n\frac{\cos\theta_i}{\cos\theta_r} = 2 - 4\frac{n\cos\theta_i}{\sqrt{1 - n^2\sin^2\theta_i}} = 0$$

solving:

$$4n^2\cos^2\theta_i = 1 - n^2\sin^2\theta_i = 1 - n^2 + n^2\cos^2\theta_i \therefore \cos\theta_i = \sqrt{\frac{1-n^2}{3n^2}}$$

which is undefined ($n > 1$). So there is no first-order bubblebow either.



Graph looks like this.

7. (a) Apply Gaussian surface, a cylinder of radius r and length l. Gauss Law:

$$E_r \cdot 2\pi r l = \begin{cases} \dfrac{\lambda l}{\epsilon_0}, & a < r < 4a \\ 0, & r \leq a \text{ or } r \geq 4a \end{cases}$$

So

$$\vec{E} = \begin{cases} \dfrac{\lambda}{2\pi r \epsilon_0}\hat{r}, & a < r < 4a \\ \vec{0}, & r \leq a \text{ or } r \geq 4a \end{cases}$$

Where $\hat{r}$ is unit vector in radial direction.

(b) Apply Amperian loop, a rectangle with one edge a distance r from the axis and the other edge limit to infinity. Let this edge have length l. The other pair of edges are along radial lines. Applying Ampere's Law:

$$B_z \cdot l = \begin{cases} \dfrac{\mu_0 \lambda l \omega}{2\pi}, & r < a \\ 0, & r \geq a \end{cases}$$

So

$$\vec{B} = \begin{cases} \dfrac{\mu_0 \lambda \omega}{2\pi}\hat{z}, & r < a \\ \vec{0}, & r \geq a \end{cases}$$

Where $\hat{z}$ is the unit vector in same direction as $\vec{\omega}$.

(c) Seeing that $\vec{B} = \vec{0}$ for $r \geq a$ and $\vec{E} = \vec{0}$ for $r < a$, the Poynting vector is $\vec{0}$ for all regions. This is true because of the non-relativistic assumption $\omega \ll \frac{c}{a}$. In fact, taking into account relativistic considerations, the charges on the rotating inner cylinder are accelerating, so they will radiate electromagnetic energy (and the Poynting vector will be nonzero).

8. In lab frame, we consider that at time t the velocity of the ship is v, while after time $t + dt$ the velocity of the ship is $v + dv$.

APhO 2014 (unrelated, go to next page for start of proof)

Shifting to the instantaneous rest frame of the spaceship: at time t' the velocity of the ship is 0, after time $t' + dt'$ the velocity of the ship is

$$dv' = \frac{(v + dv) - v}{1 - \dfrac{(v + dv)v}{c^2}} = \gamma^2 dv$$

where velocity addition is applied. We also know that by time dilation, $dt = \gamma dt'$. So putting everything together, the acceleration in lab frame is

$$\frac{dv}{dt} = \frac{dv'/\gamma\text{^}2}{\gamma dt'} = \frac{1}{\gamma^3}\frac{dv'}{dt'} = \frac{a}{\gamma^3}$$

Usually, we could proceed to solve the differential equation to find the equation of motion of the spaceship in the lab frame. But the question asks something different: it wants us to relate the relative speed to t' instead!

Proof of time dilation: (which is not necessary to show, but good for your heart)

$$ct' = -\gamma\beta x + \gamma ct$$

Differentiation both sides w.r.t to t, we have:

$$c\frac{dt'}{dt} = -\gamma\beta(\beta c) + \gamma c$$

$$\frac{dt'}{dt} = -\gamma\beta^2 + \gamma = \gamma(1-\beta^2) = \frac{1}{\gamma}$$

Hence $dt = \gamma dt'$.

(a) We apply the concept of velocity addition: after time $t' + dt'$ the velocity of the ship is $dv = adt'$, so new velocity

$$v + dv = \frac{v + adt'}{1 + \frac{vadt'}{c^2}} \approx (v + adt')\left(1 - \frac{vadt'}{c^2}\right) \approx v + \left(1 - \frac{v^2}{c^2}\right)adt'$$

$$\therefore dv = \left(1 - \frac{v^2}{c^2}\right)adt'$$

This could also be obtained from the initial part

$$dv = \frac{dv'}{\gamma^2} = \left(1 - \frac{v^2}{c^2}\right)adt'$$

Which is the same expression! We expect this to be so (both parties should be observing the same relative speed of each other). Continuing derivation:

$$at' = \int_0^v \frac{dv}{\left(1 - \frac{v^2}{c^2}\right)} = \frac{c}{2}\ln\frac{c+v}{c-v}$$

Thus

$$\frac{c+v}{c-v} = e^{\frac{2at'}{c}} \therefore v = c\frac{e^{\frac{2at'}{c}} - 1}{e^{\frac{2at'}{c}} + 1} = \boldsymbol{c \cdot \tanh\frac{at'}{c}}$$

In the limit $at' \ll c$, we have $\frac{at'}{c} \ll 1 \therefore \tanh\frac{at'}{c} \approx \frac{at'}{c} \therefore v \approx at'$ which is expected as non-relativistic constant acceleration.

(b)

$$dt = \gamma dt' = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}dt' = \frac{1}{\sqrt{1 - \tanh^2\frac{at'}{c}}}dt' = \cosh\frac{at'}{c}dt'$$

Thus by integrating:

$$\boldsymbol{t = \frac{c}{a}\sinh\frac{at'}{c}}$$

For very late times $at' \gg c$, the hyperbolic sine can be approximated to be

$$t \approx \frac{c}{2a}e^{at'/c}$$

which shows that eventually, t is exponential in t' (due to the cumulative time dilation effect as the relative speed of the lab and spaceship gets larger and larger over acceleration). This question can also be tackled by four vectors.

9. (a) Surface tension: $\boldsymbol{P_e = P_{He} + \frac{2\sigma}{R}}$ (derivation for surface tension pressure: consider small square patch. Force along a length dl on the surface is σdl)

(b) Use ideal-gas law: $E_K = \frac{3}{2}k_B T = \frac{3}{2}P_e V \therefore P_e = \frac{2E_K}{3V} = \boldsymbol{\frac{E_K}{2\pi R^3}}$
Alternatively: by De-Broglie's, $p \propto \lambda^{-1} \propto R^{-1} \therefore E_K \propto p^2 \propto R^{-2}$

$$\therefore \frac{dE_K}{E_K} = -2\frac{dR}{R}$$

But $dE_K = -P_e dV = -P_e \cdot 4\pi R^2 dR$; thus

$$\boldsymbol{E_K = 2\pi R^3 P_e}$$

Both methods are interesting, the main concept is that random quantum things behave like gases. Though if you got the answer (by dimensional analysis/any heuristic argument) but with a different numerical factor, still okay.

(c) The momentum vector has magnitude $\sqrt{2mE_K}$ and direction random (this is forced by symmetry). Thus $\overline{p_x^2} = \frac{1}{3}\left(\overline{p_x^2} + \overline{p_y^2} + \overline{p_z^2}\right) = \frac{1}{3}\left(\overline{p_x^2 + p_y^2 + p_z^2}\right) = \frac{1}{3}\overline{p^2} = \frac{1}{3}2mE_K$ and $\overline{p_x} = 0$ (both used symmetry), thus

$$\Delta p_x = \sigma_{p_x} = \sqrt{\frac{2}{3}mE_K}$$

Meanwhile, the position vector has magnitude $\leq R$ and direction random. Thus $\overline{q_x^2} = \frac{1}{3}\left(\overline{q_x^2} + \overline{q_y^2} + \overline{q_z^2}\right) = \frac{1}{3}\left(\overline{q_x^2 + q_y^2 + q_z^2}\right) = \frac{1}{3}\overline{q^2} \leq \frac{R^2}{3}$ and $\overline{q_x} = 0$ (both used symmetry), thus

$$\Delta q_x = \sigma_{q_x} \leq \frac{R}{\sqrt{3}}$$

Equality when the electron's position probability distribution is fully distributed over the surface of the sphere, not in the interior.

$$\Delta p_x \Delta q_x \geq \frac{h}{4\pi}$$

Thus

$$\sqrt{\frac{2}{3}mE_K} \cdot \frac{R}{\sqrt{3}} \geq \frac{h}{4\pi} \therefore E_K \geq E_0 = \boldsymbol{\frac{9h^2}{32\pi^2 mR^2}}$$

Again, the question asks for an estimate, so the factor may be off abit from and assumptions made (for instance, if you assumed the electron's position to be uniformly distributed within the sphere, you'll get a factor $\frac{5}{3}$ different.

(d) Finish line, substituting in a, b and c results:

$$\frac{2\sigma}{R_e} = P_e = \frac{\frac{9h^2}{32\pi^2 m{R_e}^2}}{2\pi {R_e}^3}\frac{9h^2}{64\pi^3 m{R_e}^5} = \therefore R_e = \sqrt[4]{\frac{9h^2}{128\pi^2 m\sigma}} = \boldsymbol{1.31}\ \textbf{nm}$$

Again, final answer will differ by a factor if previous equations have different factors.